

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

	Understanding of Concepts (25%)	Experimental Design (30%)	Use of Tools (30%)	Analysis of Results (15%)	Documentation (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

EXPERIMENT 1: Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

Title:
LAN Communication Using Switch and Analysis of Ethernet Frame

Aim:
To design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer, configure suitable IP addresses, verify connectivity using the ping command, and analyze Ethernet frame fields using Wireshark.

- Software Required:**
- Cisco Packet Tracer
 - Wireshark

Network Topology:
PC0 → Switch → PC1

IP Address Configuration:

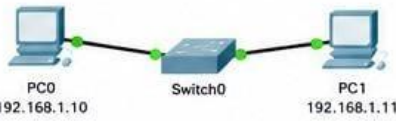
Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

- Procedure:**
1. Open Cisco Packet Tracer.
 2. Place two PCs and one switch in the workspace.
 3. Connect PC0 and PC1 to the switch using Copper Straight-Through cables.
 4. Configure PC0 with IP address **192.168.1.10** and subnet mask **255.255.255.0**.
 5. Configure PC1 with IP address **192.168.1.20** and subnet mask **255.255.255.0**.
 6. Open the Command Prompt of PC0.
 7. Execute the command: ping 192.168.1.20
 8. Verify that the ping receives successful replies.
 9. Open Wireshark and capture the network communication.
 10. Analyze the Ethernet frame generated during communication.

Ethernet Frame Fields:

Field	Description
Source MAC Address	MAC address of the sending device
Destination MAC Address	MAC address of the receiving device
EtherType	Identifies the network-layer protocol
FCS	Used for error detection

Result:



```
graph LR
    PC0[PC0  
192.168.1.10] --- Switch0[Switch0]
    Switch0 --- PC1[PC1  
192.168.1.11]
```

PC IP Configuration			
PC0		PC1	
IP Address	192.168.1.10	IP Address	192.168.1.11
Subnet Mask	255.255.255.0	Subnet Mask	255.255.255.0
Default Gateway		Default Gateway	

```
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Wireshark Capture (Filter: icmp)					
No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.1.10	192.168.1.11	ICMP	Echo (ping) request id=0x0001
2	0.000621	192.168.1.11	192.168.1.10	ICMP	Echo (ping) reply id=0x0001
3	0.001236	192.168.1.10	192.168.1.11	ICMP	Echo (ping) request id=0x0001
4	0.001824	192.168.1.11	192.168.1.10	ICMP	Echo (ping) reply id=0x0001

Ethernet II

Destination: 00:0C:29:71:1E:40
Source: 00:0C:29:3A:5F:1B
Type: 0x0800 (IPv4)
Frame Check Sequence: 0x1A2B3C4D [Correct]

Ethernet II Frame Fields	
Field	Value
Destination MAC Address	00:0C:29:71:1E:40
Source MAC Address	00:0C:29:3A:5F:1B
EtherType	0x0800 (IPv4)
FCS (Frame Check Sequence)	0x1A2B3C4D

Explanation: The Data Link layer uses MAC addresses to deliver frames within the local network.
The switch forwards the frame to the correct destination port using the destination MAC address.

The two PCs successfully communicated through the switch using the configured IP addresses. The Ethernet frame was captured using Wireshark, and the Source MAC Address, Destination MAC Address, EtherType, and FCS fields were identified.

Conclusion:

The experiment demonstrates how a switch uses **MAC addresses** to deliver Ethernet frames between devices within a local network.

EXPERIMENT 2: Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks

Title:
Communication Between Two LANs Using a Router and IPv4 Packet Analysis

Aim:
To create two LANs connected through a router, configure IPv4 addresses and default gateways, verify end-to-end communication using ping, and analyze IPv4 packet header fields using Wireshark.

- Software Required:**
- Cisco Packet Tracer
 - Wireshark

Network Topology:
PC0 → Switch0 → Router → Switch1 → PC1

IP Address Configuration:

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
Router	G0/0	192.168.1.1	255.255.255.0	—

Device	Interface	IP Address	Subnet Mask	Gateway
Router	G0/1	192.168.2.1	255.255.255.0	—
PC1	NIC	192.168.2.10	255.255.255.0	192.168.2.1

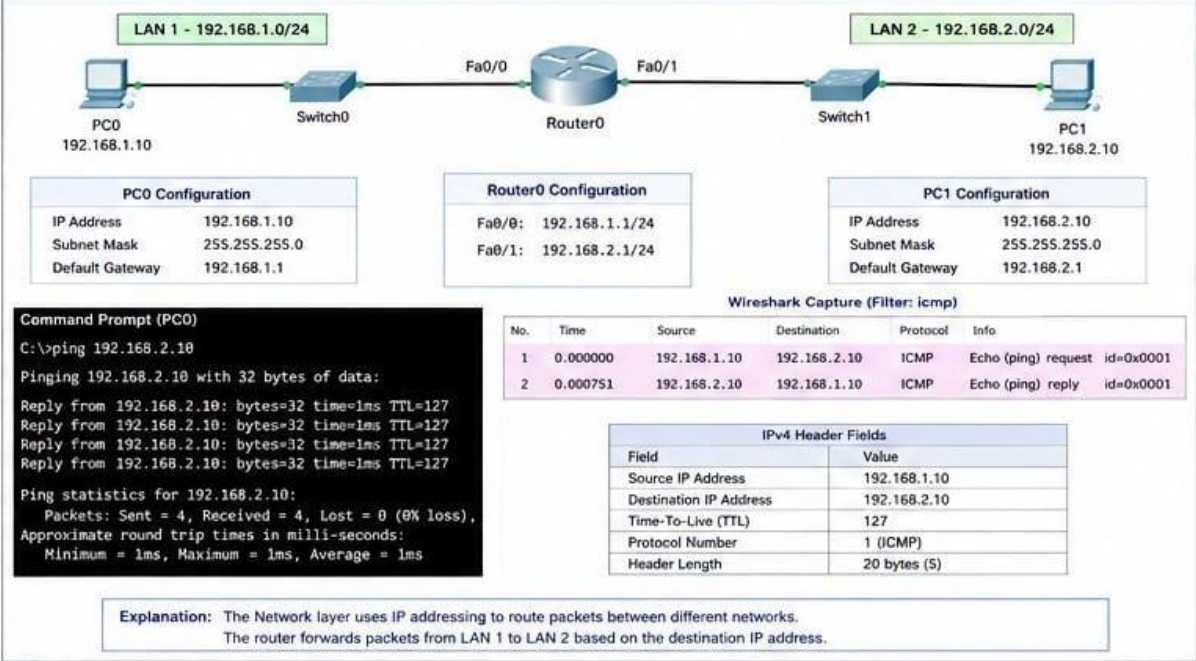
Procedure:

- 1. Place two PCs, two switches, and one router in Cisco Packet Tracer.
- 2. Connect PC0 to Switch0 and Switch0 to the router.
- 3. Connect the router to Switch1 and Switch1 to PC1.
- 4. Configure PC0 with IP address **192.168.1.10/24**.
- 5. Configure its default gateway as **192.168.1.1**.
- 6. Configure PC1 with IP address **192.168.2.10/24**.
- 7. Configure its default gateway as **192.168.2.1**.
- 8. Configure the router interfaces with **192.168.1.1** and **192.168.2.1**.
- 9. Enable the router interfaces.
- 10. From PC0, execute: ping 192.168.2.10
- 11. Verify successful communication.
- 12. Capture the ICMP packets using Wireshark.
- 13. Analyze the IPv4 header fields.

IPv4 Header Fields:

Field	Description
Source IP Address	IP address of the sending device
Destination IP Address	IP address of the receiving device
TTL	Limits the lifetime of the packet
Protocol Number	Identifies the upper-layer protocol
Header Length	Indicates the size of the IPv4 header

Result:



Successful communication was established between the two LANs through the router. The ICMP packets were captured and the IPv4 header fields were analyzed.

Conclusion:

The experiment demonstrates how the **Network layer** uses IP addressing and routing to forward packets between different networks.

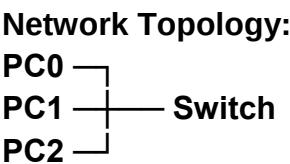
EXPERIMENT 3: Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Title:
ARP Request and Reply Analysis Using Wireshark

Aim:
To construct a LAN with multiple hosts, clear the ARP cache, initiate communication using ping, and analyze ARP Request and ARP Reply packets using Wireshark.

Software Required:

- Cisco Packet Tracer
- Wireshark



IP Address Configuration:

Device	IP Address	Subnet Mask
PC0	192.168.10.10	255.255.255.0
PC1	192.168.10.20	255.255.255.0
PC2	192.168.10.30	255.255.255.0

- Procedure:**
1. Open Cisco Packet Tracer.
 2. Add three PCs and one switch.
 3. Connect all PCs to the switch.
 4. Configure the IP addresses.
 5. Clear the ARP cache of PC0.
 6. Initiate communication from PC0 to PC1 using: ping 192.168.10.20
 7. Capture the communication using Wireshark.
 8. Identify the ARP Request packet.
 9. Identify the ARP Reply packet.
 10. Analyze the sender and target MAC and IP addresses.

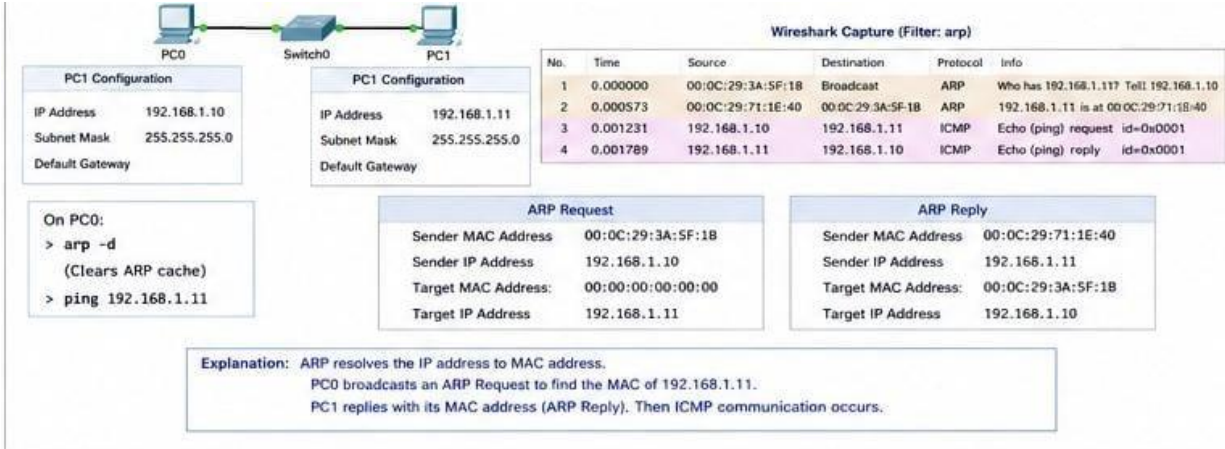
ARP Analysis:
ARP Request

The sender asks:
“Who has 192.168.10.20?”
The request is sent as a broadcast because the sender does not initially know the destination MAC address.

ARP Reply:

PC1 responds with its MAC address, allowing PC0 to associate:
192.168.10.20 → PC1's MAC Address

Result:



The ARP Request and ARP Reply packets were successfully captured and analyzed using Wireshark.

Conclusion:

ARP resolves a logical **IPv4 address** into a **physical MAC address**, allowing Ethernet communication to take place within the local network.

EXPERIMENT 4: Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Create a client-server network

Title:
ICMP Echo Request and Echo Reply Analysis

Aim:
To develop a network consisting of two PCs connected through a router, verify connectivity using ping, capture ICMP traffic using Wireshark, analyze ICMP fields, and calculate Round Trip Time.

- Software Required:**
- Cisco Packet Tracer
 - Wireshark

Network Topology:
PC0 → Router → PC1

IP Address Configuration:

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
Router	192.168.1.1	255.255.255.0	—

Device	IP Address	Subnet Mask	Gateway
Router	192.168.2.1	255.255.255.0	—
PC1	192.168.2.10	255.255.255.0	192.168.2.1

Procedure:

- 1. Place two PCs and one router in Cisco Packet Tracer.
- 2. Connect PC0 and PC1 to the router.
- 3. Configure the IP addresses and subnet masks.
- 4. Configure the default gateways.
- 5. Enable the router interfaces.
- 6. From PC0, execute: ping 192.168.2.10
- 7. Verify that Echo Reply messages are received.
- 8. Capture the ICMP packets using Wireshark.
- 9. Analyze the ICMP Echo Request and Echo Reply packets.
- 10. Record the request and reply timestamps.
- 11. Calculate the Round Trip Time.

ICMP Fields:

Field	Description
Type	Identifies the ICMP message type
Code	Provides additional information about the message
Identifier	Identifies the ICMP request

Sequence Number Identifies the order of requests

RTT Calculation:

RTT = Time of Echo Reply – Time of Echo Request

For example, if:

Echo Request = **1.250000 seconds**

Echo Reply = **1.253000 seconds**

Then:

RTT = 1.253000 – 1.250000

RTT = 0.003 seconds = 3 ms

Result:

PC0 Configuration

IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

PC1 Configuration

IP Address	192.168.2.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1

Router0

PC0 --- Fa0/0 --- Router0 --- Fa0/1 --- PC1

```
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=2ms TTL=127
Reply from 192.168.2.10: bytes=32 time=2ms TTL=127
Reply from 192.168.2.10: bytes=32 time=2ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

Wireshark Capture (Filter: icmp)

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.1.10	192.168.2.10	ICMP	Echo (ping) request
2	0.001543	192.168.2.10	192.168.1.10	ICMP	Echo (ping) reply

ICMP Fields

Field	Echo Request	Echo Reply
Type	8 (Echo Request)	0 (Echo Reply)
Code	0	0
Identifier	0x1234	0x1234
Sequence Number	1	1

Explanation: ICMP Echo Request and Reply are used to test connectivity.
RTT (Round Trip Time) = Time of Reply - Time of Request = 0.001543s - 0.000000s = 0.001543s (2ms)
ICMP helps in troubleshooting and verifying network connectivity.

ICMP Echo Request and Echo Reply packets were captured and analyzed successfully. The Round Trip Time was calculated from the packet timestamps.

Conclusion:

ICMP is useful for **network troubleshooting and connectivity verification** because it provides Echo Request and Echo Reply messages for testing communication between network devices.

EXPERIMENT 5: Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

Title:
HTTP Communication and TCP Three-Way Handshake Analysis

Aim:
To create a client-server network, configure an HTTP web server, access the webpage from a client, and analyze the TCP three-way handshake and HTTP communication using Wireshark.

- Software Required:**
- Cisco Packet Tracer
 - Wireshark

Network Topology:
Client PC → Switch → Web Server

IP Address Configuration:

Device	IP Address	Subnet Mask
Client PC	192.168.1.10	255.255.255.0
Web Server	192.168.1.20	255.255.255.0

- Procedure:**
1. Open Cisco Packet Tracer.
 2. Add one client PC, one switch, and one server.
 3. Connect the PC and server to the switch.
 4. Configure the client PC with IP address **192.168.1.10**.
 5. Configure the server with IP address **192.168.1.20**.
 6. Open the server configuration.
 7. Enable the **HTTP service**.
 8. Open the web browser on the client PC.
 9. Enter the server address: http://192.168.1.20
 10. Verify that the hosted webpage is displayed.
 11. Capture the communication using Wireshark.
 12. Identify the TCP three-way handshake.
 13. Identify the HTTP GET request.
 14. Identify the HTTP response.
 15. Analyze the TCP communication.

TCP Three-Way Handshake
The TCP connection is established using:

SYN → SYN-ACK → ACK:

- **SYN:** Client requests a TCP connection.
- **SYN-ACK:** Server acknowledges the request.
- **ACK:** Client confirms the connection.

After the connection is established, the client sends an:
HTTP GET request

The server responds with the requested webpage through an:
HTTP response

Result:



The client successfully accessed the webpage hosted by the server. The TCP three-way handshake, HTTP GET request, and HTTP response were captured and analyzed using Wireshark.

Conclusion:

TCP provides reliable delivery for HTTP traffic through **connection establishment, acknowledgments, and ordered data transfer.**